

Morphological Analysis and Diagnosis of Human Leukemic Cells Using Image Processing: A Review

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Abstract—Leukemia is a type of blood cancer in which abnormal and malfunctioning white blood cells (WBC) proliferate rapidly and on a large scale and hence, it must be detected and diagnosed quickly. Manual detection of leukemia is a time-consuming, labor-intensive procedure that requires the assistance of a hematologist and considerably depends on factors such as the hematologist's experience and tiredness causing the results subjected to error and untrustworthy. These issues are addressed by automating the morphological analysis of blood cells, which reduces time, cost, and human error. It does not necessitate costly lab experiments and testing, and hence images are inexpensive, it will be a better solution for hospitals and laboratories. This automation process uses microscopic images obtained from a dataset, the internet, or a lab or hospital, and it may employ many image processing techniques, and so on. Automated detection and diagnosis of leukemia emerge as a challenging area which lay the groundwork to many research opportunities. This paper to discuss on morphological analysis in terms of detection, counting and diagnosis of leukemia, a comparative review and identified research gaps on image processing techniques used in state-of-the-art research. In the review, it was revealed that, Morphological/shape-based segmentation has verified as the best method for image segmentation in ALL diagnosis and K-Nearest Neighbor, Support Vector Machines (SVM) and Ensemble of classifiers have resulted better classification results. One major finding is the dependency of datasets in the accuracy of the solutions and hence there are no standard methods and metrics to compare accuracy among various researches. This issue with the dataset throws into question whether the researchers achieved acceptable high accuracy and expected outcomes. Another dilemma is that when they are employed in real-world settings, they still require final verification from medical staff, suggesting that current developments must be fine-tuned for trustworthiness and reliability. Standardized and scrutinized further research into leukemia diagnosis would provide solid remedies to the present challenges with current image processing-based diagnosis methods.

Keywords—Automated Leukemia Detection and Diagnosis, Image Processing Techniques, Morphological analysis

I. INTRODUCTION

Blood holds a unique place in human lives, due to its existence as the only fluid tissue and as well as a most vital component of the human body being the primary factor that allows humans to live. Human blood is made up of stem cells, which are a type of specialized cell found in the bone marrow. Bone marrow is a spongy substance found in the body's major organs. It is well known that blood cells mainly include Red

blood cells (RBC), White blood cells (WBC), platelets and plasma. WBC is of five types as Neutrophils (50-70%), Eosinophils (3-4%), Basophils (0.51%), Monocytes (3-8%), and Lymphocytes (25-45%). Neutrophils, Eosinophils and Basophils have granules in their cytoplasm; hence they are named as granulocytes. Monocytes and Lymphocytes do not have granules, thus are termed agranulocytes [1]. Leukemia is a blood cancer in which number of WBC increase, and they are immature cells that malfunctions destroy other cells. The ratio between red cells and white cells in blood is 1000:1. If an increase of the number of WBC in numerous way remarkably occurred and the size of them have been changed, then that person is succumbed to suffer from leukemia [1, 2].

Leukemia is the most prevalent type of blood cancer across all age categories, particularly in children. This abnormal phenomenon is produced by excessive blood cell proliferation and immature growth, causing extensive harm to red blood cells, bone marrow, and the immune system. These malignant white blood cells, or lymphoblast, even spread to other organs like the spleen, brain, liver, and kidneys, where they metastasis to vital tissues in human body.

There are two types of leukemia based on the white blood cell from which the cancer originates from. Lymphocytic leukemia is caused by abnormal lymphocytes, which are known as lymphoblasts and Myelogenous leukemia is caused by abnormal monocytes known as myeloid or nonlymphocytic. Leukemia can also be grouped in to two sub areas based on how quickly it proliferates and become severe. Acute leukemia involves the rapid overgrowth of very immature blood cells whereas chronic leukemia involves the overgrowth of mature blood cells and is a bit slow comparing with the acute. Accordingly, the four major categories of leukemia can be named as Acute lymphoblastic leukemia (ALL), Acute myelogenous leukemia (AML or Myeloblastic), Chronic lymphocytic leukemia (CLL) and Chronic Myelogenous leukemia (CML) [2,3].

Acute Lymphoblastic Leukemia (ALL) is the most common leukemia type in young children and adults, especially over 60 and older. The survival rate in children is 85% and 50% in adults. Chemotherapy and radiotherapy are the well-known standard treatments for ALL. Subtypes of ALL include precursor T acute lymphoblastic leukemia, precursor B acute lymphoblastic leukemia, acute biphenotypic leukemia and Burkitt's leukemia [4]. Chronic Lymphocytic Leukemia (CLL) is the most common type in adults over 55 years and sometimes

effect in younger adults. CLL never affects children. The five-year survival rate of this is 75%. Even though there are many succeeded treatments, this is an aggressive cancer that is incurable. B-cell prolymphocytic leukemia is a common subtype of CLL [4]. Acute Myelogenous / Myeloblastic Leukemia (AML) is mostly affected in adults than in children and it's more common in men than women. Chemotherapy is used for treatment. Acute promyelocytic leukemia (APL), acute megakaryoblastic leukemia and acute myeloblastic leukemia are the subtypes of AML. The five-year survival rate of AML is 40%, apart from APL which is over 90% [1,2,3]. Chronic Myelogenous Leukemia (CML) affects adults and a small percent in children. Chronic myelomonocytic leukemia is a one subtype of CML [2,3].

A regular blood test that results in an abnormal blood cell count is usually used to diagnose leukemia. When leukemia is suspected based on blood testing, the doctor examines blood and bone marrow samples to study the cell morphologies. The same samples are also sent to the pathology laboratory to identify proteins on the surface and chromosomal abnormalities. This is the current criterion for diagnosing individual leukemia [5].

Existing Methods for Leukemia Diagnosis includes:

- Medical history and physical examination.
- Complete Blood Count (CBC) & Differential blood count (DBC): Blood is checked under the microscope for the count of RBCs, WBCs and platelets. CBS is performed using hemocytometers and DBC is a manual method done using a microscope [6].
- Bone marrow aspiration: A Bone marrow sample is obtained from the breastbone with the help of a needle and subjected to microscopic observation observing abnormal cells.
- Cytogenetic analysis: Test blood and bone marrow to identify individual chromosomes. Usually takes about 3 weeks for the results and they show the abnormalities in chromosomes that help in diagnosis and identify the type of leukemia.
- Immunohistochemistry: Blood samples are treated with special antibodies in immunohistochemistry, and they are observed under the microscope to view the colour changes. The results aid to determine the leukemia type and examine the blast cells.

Recent advances in Image Processing, has provide ample inventions in biomedical imaging systems such as Medical image management and image data mining, Blood group typing and Blood phenotyping [7], Bioimaging, Virtual reality in medical visualization and Neuroimaging and there are some challenging issues in these areas too [8]. These Biomedical imaging systems has been able to provide excellent methodologies for detection and counting of leukemia in blood samples [9].

Section "State of The Art Techniques Used in Morphological Analysis" of the paper discusses early research attempts. Section. "Research Findings and Discussion" is followed by the main findings of the study. Section "Conclusion" concludes by indicating the research challenges

for morphological image analysis for leukemia detection and diagnosis and highlight the feasibility of Image processing for a better solution.

II. STATE OF THE ART TECHNIQUES USED IN MORPHOLOGICAL ANALYSIS

The common flow of the image processing techniques that have been used in diagnosis of leukemia in existing literature involves major five steps as: Image acquisition, Image pre-processing, Image segmentation, Feature extraction and Classification. A massive contribution has been done by researchers in the aim of leukemia detection and counting the cells in image processing at present. This section would first cover an insight to image databases used and then present an overview of previous research done on image based morphological analysis in leukemia diagnosis.

A. Leukemia Image Databases

1) *ALL-IDB Image Database*: Acute Lymphoblastic Leukemia Image Database (ALL-IDB) is a public dataset of blood samples that is specifically developed for the evaluation and the comparison of algorithms in segmentation and classifications in automatic acute leukemia detection [10]. The founders of ALL-IDB presents morphological variability in the blast cells of ALL patients & non-ALL that has been classified according to the FAB (French-American-British) classification which is the widely used leukemia classification by morphological analysis. According to the FAB method, the candidate lymphoblasts are classified into three groups as L1, L2 and L3 based on the size of the blasts, cytoplasm, nucleus and vacuoles. The ALL-IDB database has two versions as ALL-IDB1 and ALL-IDB2 which are freely available online. Figure 1 depicts sample blood image retrieved from ALL-IDB database which illustrate morphological differences between healthy and blood with ALL blasts.

2) *BloodSeg*: "BloodSeg" [11] comprises 367 blood smear photos with a resolution of 640 x 480 pixels that were captured with an optical microscope and a color CCD camera. The microscope magnification is 100 objective of the lens. An expert hematologist has performed ground truth nucleus segmentation on these images.

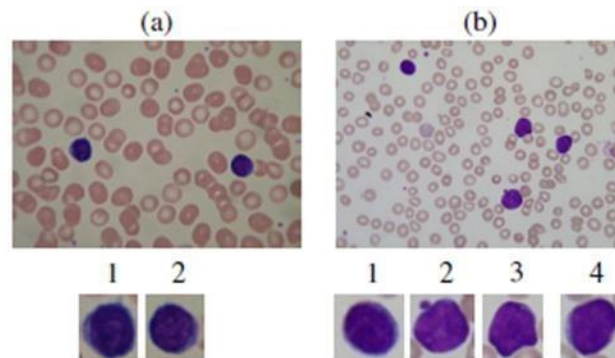


Fig. 1. Examples of blood images: (a) healthy blood, (b) Blood with ALL blasts. (a1-2) and (b1-4) are zoomed subplots of the (a) and (b) images centred on lymphocytes and lymphoblasts. Source. ALL-IDB [10]

3) *CellaVision*: This database has 100 photos with 300x300 pixel resolution. A hematology specialist has performed ground truth segmentation of the nucleus and cytoplasm in this database as well.

4) *LISC: Leukocyte Images for Segmentation and Classification*: This database contains hematological pictures taken from 400 samples that were acquired from 100 microscope slides of 8 healthy patients' peripheral blood. The microscope slides have been smeared and stained using the Gismo-Right process, and images were captured using an achromatic lens at a magnification of 100 on a light microscope (Microscope-Axioscope 40) from the stained peripheral blood. These photographs have then been captured with a digital camera (Sony Model No. SSCDC50AP) and saved in BMP format. The photos have a total resolution of 720 x 576 pixels [12].

There are several other benchmark image datasets publicly available that have been used by early researches like; HistologyDS2828, LYMPH, leukocytes and MLL_leukemia etc. [13]. Many researchers have conducted research on leukemia detection based on various methods over past years. This section is followed by summarizing the main techniques used in former research.

B. Fuzzy Methods and Watershed Transform

Bhattacharjee and Saini [14] have presented a robust technique for leukemia detection and a GUI that is developed for segmentation and classification of lymphocytic cells in recognition of acute lymphoblastic leukemia. The researchers have retrieved input images from the ALL-IDB database. The developers have used morphological operators' segmentation and have given a comparison of segmentation accuracy of it with other existing methods, Watershed transform and HSV colour model, Fuzzy C Means Clustering, Fuzzy K-Means Clustering in $L^*A^*B^*$ colour space, Lindblad Distance transform and Otsu's thresholding. In that it has been discovered that the morphological operators used for the extraction of features have resulted a high segmentation accuracy. A comparison between performance accuracy of various classification models used also have given. It has been discovered that KNN is the best classifier that produced high specificity and have the lowest computational complexity. The GUI have ability to browse, save, restore, and store images, image filling and many other features. It also performs important functions auto check and manual check that counts the RBC and WBC and determine the types of the WBC.

Viswanathan [15] have proposed fuzzy C means-based detection based on morphological contour segmentation. This approach has been presented under three consequential steps as Nuclei separation, Morphological Contour Segmentation of Nuclei and Fuzzy C mean Detection of Leukemia. In step one contrast stretching, histogram equalization, addition and subtraction processes have done. In step two, the segmentation has been based on local and global curvature properties. The Fuzzy C mean clustering algorithm has been used to cluster the feature points to the degree specified by a membership grade. As per the future work, they propose to extend their solution with soft set rules in further work to classify the types of cancer cells affected in WBCs.

Mohapatra and the colleagues [16] is another research group who have used fuzzy based blood image segmentation in research. In their project, a fuzzy based two stage color segmentation strategy has been employed for separate out leucocytes from other blood components. Blood segmentation has been achieved by the fuzzy clustering technique, Gustafson Kessel clustering, and it has followed by nearest neighbour classification in $L^*a^*b^*$ space. Sub-imaging was done through bounding box technique. Fractal dimension, contour signature and shape feature extraction has been performed in feature extraction and discriminative features such as nucleus shape, texture are used for final detection of leukemia.

C. Neural Network Classification

Mohapatra and el have engaged in another project in Lymphocyte image segmentation using Functional Link Neural Architecture for ALL detection and diagnosis [17]. In that research image segmentation has been deliberated as a pixel classification problem and a neural architecture has been employed for classification. Classification of lymphoblasts into its subtypes and considering touching cells in image segmentation and leukemia type classification are included as their further work of this research group.

Scotti and Piuri, who were major contributors of ALL-IDB had conducted research to analyze the morphological classification of Leucocytes by microscopic images [18]. In this paper, the researchers have focused on reducing the problem of identification and classification of WBC types in microscope images. In their experiments, they have considered a parallel classifier built with feed-forward neural networks. They have used the Levenberg Marquard training method along with Bayesian regularization available in the Matlab Neural Network Toolbox. The results shown the parallel classification system has reached the lowest mean miss-classification error.

Scotti has later conducted a research on Automatic Morphological Analysis for Acute Leukemia Identification based on peripheral blood microscope images [19]. The researcher presents a structure of the modules composing the ALL detection system comprised of five main steps as single-cell selector, white cell selector, lymphocyte selector, feature extraction and cell classifier. Among these the last two methods have been explained as the proposed system in this paper. The first three methods have been discussed in the previous research paper [18]. The morphological indexes outputted by the feature extraction module has been the input for classifier module which classifies whether a cell is blast or normal. The feature extraction module has performed in three sub phases as membrane selection, cytoplasm selection and nucleus selection [19].

Fatma and Sharma [20] have tried on a system to identify and classify sub types of acute leukemia using neural network. Linear contract enhancement, HIS Colour Modelling, K-means clustering, Median filter have been effectively performed. The statistical and colour based features obtained via feature extraction had been considered as an input dataset for the classification by feed-forward multilayer neural network. As further work, the group has recommended larger datasets have to be used and suggest on trying it with Support Vector Machines for better accuracy in classification.

D. Morphological Shape-Based Analysis

Segmentation based on morphological operations is a strategy that considers the processing of geometrical structure based on set hypothesis, topology, lattice hypothesis, and arbitrary functions, among other variables. This is the most successful segmentation approach employed to date.

Vaghela and the group [21] have presented various segmentation approaches like histogram equalization, linear contrast stretching, some morphological techniques like area opening, area closing, erosion, dilation, Watershed transform, K means, histogram equalization & linear contrast stretching, and shape-based features that have been used by early researchers in leukemia detection. The researcher presents limitations of pre-mentioned techniques and the advantage/merits of shape-based features over them. Thus, shape-based method based on statistics and its capability of easy detection of white and overlapping cell and shape of cell, it has been verified as the best method for segmentation.

Patel and Mishra [22] have discovered an automatic approach for leukemia detection using microscopic images. Median filter has been used for noise removal and Wiener filter for blurriness removal. Roundness measure has been used to discover the grouped leucocytes. Cseke's observation has been used in finding the nucleus in this development. Colour, geometric, shape and statistical features have been analyzed and classified under the SVM classifier in the intention of grouping the normal and abnormal cells.

The robust feature extraction and selection technique for the recognition of lymphocytes versus acute lymphoblastic leukemia has been developed by researchers Madhlom and the group [23]. The group have considered shape features and two types of texture features as first-order texture gained through histogram statistic and second-order texture that is based on the gray level co-occurrence matrix. From a selected group of 30 features, the best features have been chosen through a specific process carried out consisting of normalization and ranking the features using scalar feature selection method with a Fisher's Discrimination Ratio (FDR). Then, exhaustive search has been used to nominate the best three features. As per the further work, they suggest this approach can also be expanded to test classification accuracy of ALL and AML.

The above algorithm proposed by Madhlom and the group has been modified to reduce noise and enhance the accuracy by Mohamed and the group which is another group of researchers [23]. To overcome the limitations of Madhlom concept, they have proposed to use the minimum segment size of the nucleus as a constraint to remove the non-nucleus objects. It is chosen to be half of the RBC average size. The segmentation accuracy has been comparatively calculated for the five types of WBC separately by the new algorithm and Madhlom concept. According to the results, Monocytes have shown the highest accuracy, Eosinophils have the lowest segmentation accuracy and for the rest of cell types, the performance is around 80%. The researchers have compared their algorithm with the Madhlom et al. The proposed algorithm has shown 79.7% accuracy compared to 55.9% of Madhlom.

E. Ensemble Classifier Systems

Mohapatra and group is a prominent research group who researched on using ensemble classifier system for early diagnosis of ALL [24]. The classifiers they have used include KNN, NB, RBFN, MLP, SVM and EOC (Ensemble of classifiers). As the results they have obtained more accuracy in EOC in comparison with other classifiers employed. In this project, Lymphocyte image segmentation has done using shadowed C-means clustering (SCM) performed in L^*a^*b (CIELAB) colour space. 44 features considering shape, texture and colour have been collected in feature extraction. One other important fact in this research compared to other research is they have performed Data normalization prior to the classification. At the end performance analysis has been done by introducing a confusion matrix.

Automatic leukemia detector is a software developed using LabVIEW and MATLAB for leukemia detection [4]. The researchers have formed two separate algorithms require for the nucleus segmentation in each of the two software. In each of these algorithms, Contrast stretching techniques such as, linear contrast stretching, and histogram equalization have been combined with image arithmetic operations to remove RBCs and other fragments of the WBC and retain only the nucleus of the WBC. Moreover, the fully segmented nucleus had been able to be better achieved by using LabVIEW based algorithm since it is less sensitive to input image variations.

F. Clustering Approaches

C-Means, K-Means, K-Nearest Neighbour (KNN) clustering are among the popular clustering mechanisms used in leukemia diagnosis [22,23,24]. A three-layered framework consists of feature extraction, coding, and classification for the detection of leukemia from blood smear images has been proposed by Faivdullah and his colleagues [25]. They have employed dense Scale-Invariant Feature Transform (DSIFT) for Feature extraction and have used 16 histograms to store geometric data of images. In coding process, a codebook that has been created using 1500 of vectors using k-means clustering was used. That coded feature vector has then been subjected to SVM to classify leukemia types. They have employed a one-vs-all technique to convert SVM, which is a binary classifier into a multi-class classifier.

K-means clustering technique has been applied for image segmentation by Vaghela and the group [21]. Process flow used to separate white cell and red cell using K-means clustering method by them is shown in figure 2 that is. They have achieved 72% accuracy and have identified key points as K-means does not give accurate classification results with labeled data and not applicable on incremental data as well.

Shyalika and el. [26,27] is another group who have used morphological based image segmentation and KNN clustering in leukemia diagnosis. They have discovered that the morphological operators used for the extraction of features have resulted in high segmentation accuracy. They have extended their development to solve the issue of overlapping cells in leukemia detection and diagnosis, where Grab cut algorithm and circle detection algorithm has been employed to cater the issue with cell overlapping in liquid-based blood layers.

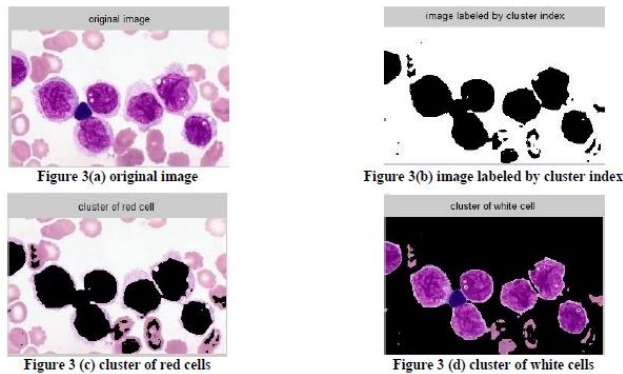


Fig. 2. Process Flow for K means Clustering used by Vaghela and el. [21]

G. CART (Classification and Regression Trees)

CART (Classification and Regression Trees) statistical software is an attempt in automatic leukemia diagnosis that investigates the classification power of cell markers extracted in segmentation [28]. This software generates classification tree diagrams with complete splitting information at each node. Then it produces a classification matrix, splitting cost and probability matrix for both the learning sample and the cross validation. The classification trees can be saved and used in classifying unknown specimens.

Lin and Hong [29] have invented an algorithm for differential count of WBC in noisy normal blood smear images. In this paper, the researchers have presented a methodology to ease the effect of un-uniform stain and extract the WBC nucleus for further analysis. They have calculated the number of lobes in cytoplasm to identify granulocytes and agranulocytes separately. In segmentation, a threshold for S-channel has set to get the seed for growing nucleus region and region growth has been done to get the proper nucleus regions.

H. Local Binary Pattern (LBP)

Local Binary Pattern (LBP) is a texture feature descriptor used in automatic detection of acute leukemia. Singhal and Singh were the first group of researchers to use LBP for leukemia detection [30]. Texture provides statistics on the spatial arrangement of intensities in an image. LBP is a computationally simple texture feature that is robust to deviations in intensities. Leukemia causes major changes in the chromatin distribution of lymphocyte nucleus that it can be figured in the form of texture [30]. LBP has been computed to check these texture variants in lymphocyte nucleus image. The researchers have used images from ALL-IDB2 database [10,30] in the system. They have extracted geometric features from the lymphocyte and have compared classification results for both types of features. As per the results, LBP features has performed well as compared to geometric features.

Agaian et al. is a major group that conducts research especially on detection of Acute Myelogenous Leukemia (AML) [31]. They have developed an automatic screening system for AML detection in blood smear. They have converted the RGB image into CIELAB in pre-processing and for segmentation k-mean clustering algorithm is used. In feature extraction, Box counting algorithm has been used in Hausdorff dimension. It is a widely used method in medicine and science

fields for various quantitative measurements. Gray level cooccurrence of matrix (GLCM) is used for finding the shape and texture parameters of nucleus and Local Binary Pattern was used in texture classification. They have undergone in two phases of implementation, training phase and testing phase during the final execution of the system.

I. Hough Transform

Maitra and the team [32] have discovered a technique for the detection and counting of RBC in blood sample images using Hough Transform. Hough Transform which was been initially used for detecting circles and lines in edge maps of images had been employed in finding the RBC based on their sizes and shapes. Hough Transform has been used as a voting procedure to find imperfect instances of objects in a certain class of shapes. The researchers have invented a formula to count the actual RBC Count per cumm based on the count gained by Hough transform, input image area, magnification, and film thickness and dilution factor. The evaluation results have shown that an enhanced conformity with the manual counting method had been achieved by the proposed method. The group finally conclude that their software must be modified to count the RBCs that are partially in the image field, complete blood cell count, white blood cells and platelets in the blood sample.

J. Segmentation Using HSV Colour Space

Automatic blast counting system for acute leukemia based on blood samples has been developed by Halim and his colleagues [33]. Segmentation using HSV colour space has been done. Out of the three colour spaces, Hue, saturation, and value, only the S component has been used in this scenario for transforming the RGB image. Among the two mathematical operators in mathematical morphology, dilation, and erosion, only erosion has been implemented in this project to remove small objects. The counting process has been accomplished using the BWLABEL function of Matlab 7.0. The same researchers have engaged in introducing an efficient nucleus segmentation methodology for acute leukemia [33]. This discusses about the application of Global Contrast Stretching (GCS) and segmentation based on HSI (Hue, Saturation, Intensity) colour space that has been used to improve the image quality. GCS is generally performed by applying a sliding window (known as the KERNEL) across an image and then, adjusting the center element using the formula. A comparison between the original images and after enhancement has been done by applying GCS for both ALL and AML cells. The outcomes of the research show that the proposed enhancement procedure is useful to extract the nucleus in WBC images of both ALL and AML images by using the same threshold value. Therefore, the researchers propose that this methodology can be effectively used to contrast between AML and ALL [33].

K. Sri Lankan Case Studies on Morphological Analysis

Among the Sri Lankan research conducted in this field, the project "Lohitha" conducted by researchers of University of Moratuwa is a related research [34]. It mainly analyzed RBC, WBC and platelets of blood sample, but did not discovered the existence of abnormalities. A team of researchers from Sri Lanka Institute of Information Technology has conducted a project which focused mainly on identifying red blood cell abnormalities. The aim of the project was to support the

diagnosis of indicative diseases. The group identified and counted the presence of Sickie cells, Pencil cells, Stomatocytes, Macrocytes, Microcytes, Fragmented cells and Target cells of RBC and this system produces three reports as Image Full Blood Report, Image Differential Blood Report, and Image Abnormal Cell Report [34].

III. RESEARCH FINDINGS AND DISCUSSION

This section provides a comparative analysis of the Image processing and machine learning techniques used in early studies in leukemia diagnosis. Table I highlights the merits and demerits summarized in the research study. When comparing the accuracy, ensemble classifier systems and Morphological shape-based analysis have produced considerably better results in leukemia detection and diagnosis than other methods. Most of the studies used the ALL-IDB Image Database, while others used databases such as BloodSeg, CellaVision, and others, and a few used custom databases of live images. Majority of the research applications in this regard have been implemented with the tools MATLAB, Python, and LabVIEW.

TABLE I. COMPARATIVE REVIEW ON IMAGE PROCESSING TECHNIQUES USED IN LEUKEMIA DIAGNOSIS

Method and paper reference	Merits	Demerits
Watershed transform [14]	Considerably easy to detect of white cells	Cannot expect high accuracy and cannot implement on every blood image
Clustering based approaches [14,22,23,24,26, 27,31]	It is used for clustering and separate the data based on value of K.	It does not give classification with labeled data and not applicable on incremental data.
Edge detection using histogram equalizing method and linear contrast stretching [4 ,15, 20, 33]	This is very useful method to detect white cell and for contrast enhancement.	It is hard to define boundary of overlapping cell.
Segmentation using HSV colour space [14,33]	Increased accuracy, higher speed and advanced colour shade processing	Can cause bias due to some color spaces that perform better segmentation to a given color, or a given illumination condition
Shadowed C-means clustering(SCM) [22,23,24]	SCM can handle overlapping among clusters efficiently and can model uncertainty in Class boundaries. Robust in the presence of outliers.	Have problems with high dimensional data sets and many prototypes.
Hough Transform [32]	Able to detect lines with short breaks in them due to noises, or when objects are partially occluded.	It can give deceptive results if objects are aligned. So, detection of overlapping cells can be inaccurate.
CART (Classification and Regression Trees) [28 ,29]	Less training period. Robust to outliers and can handles non-linear parameters efficiently.	Overfitting and so may have High variance. Unstable and can be affected by noise in cell detection.

Method and paper reference	Merits	Demerits
Ensemble classifier systems [4 ,23]	Produce more accurate results.	Can lead to complexity and ensembles cost more to create, train, and deploy.
Morphological shape-based analysis [21,22,26,27,34]	Very easy for the detection of white and overlapping cell and shape of cell	This is based on statistics so can get approximate result.

It is obvious that many attempts have been made in recent years to analyze and discover novel approaches in the recognition, diagnosis and classification of leukemia using images from benchmark datasets. To demonstrate accuracies, researchers must now investigate new live image datasets comprised with real Peripheral blood smear (PBS). Since all the studies were not conducted on the same dataset with the standard metrics, identical images of same size and resolution, comparing the accuracy and efficiency of algorithms has been a difficult issue. This problem with the dataset even calls into question if the researchers attained provable high accuracy and predicted outcomes. Another important point of contention is that, even though the developed procedures seem performing considerably effectively at present, verification from a hematologist or pathologist is still required when they are used in real-world settings. Image processing algorithms must be carefully developed, experiment and deployed in real world, since it would ultimately lead to decide life of human beings. Thus, this research area undoubtedly is challenging and must be scrutinized handled.

IV. CONCLUSION

As per the literature survey, it was able to gather ample evidence regarding new methodologies, concepts, techniques, automatic systems developed and ongoing researches that are been conducted in detection and counting leukemia cells. Morphological/shape-based segmentation was shown to be the best approach for image segmentation in ALL diagnosis, whereas K-Nearest Neighbor, Support Vector Machines (SVM), and Ensemble of classifiers produced higher classification results. It was concluded that the current state of development has limitations and troublesome regions. One key conclusion is the reliance of various datasets on the accuracy of findings, making different studies proves to be incomparable and anticipated outcomes being questionable. Another significant point of notice is that, while the developed processes appear to be operating quite well now, verification from a hematologist or pathologist is still required when they are deployed in real-world circumstances, evidencing that current developments would still need to be fine-tuned for trustworthiness and reliability.

Therefore, we recommend further study on detection and diagnosis of leukemia using image processing and with standard and acceptable methodologies which might provide solutions for existing issues. According to statistics data presented in the future spread of a malignancy like leukemia around the world, automated processes with high accuracy, reliability and usability to diagnose leukemia have become an essential requirement. The research directions for present issues and possible solutions would pave the way for future research on image processing

applications for human disease diagnosis which would be more cost effective, applicable, and efficient to use for the benefit and longevity of mankind.

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